

Amir Asiaee^{Taheri}

Education

- 2007 **B.S. in Computer Engineering - Software**, University of Tehran, Tehran, Iran
- 2010 **M.S. in Artificial Intelligence and Robotics**, University of Tehran, Tehran, Iran
○ Thesis Topic: *Studying the Effect of Structure on Spreading Process in Complex Networks*
○ Adviser: [Professor Masoud Asadpour](#)
- 2017 **Ph.D. in Computer Science**, University of Minnesota, Minneapolis, MN, USA
○ Thesis Topic: *High Dimensional Learning with Structure Inducing Constraints and Regularizers*
○ Adviser: [Professor Arindam Banerjee](#)
- 2021 **Postdoctoral Fellow**, [Mathematical Biosciences Institute](#) and [Department of Biomedical Informatics](#), The Ohio State University, Columbus, OH, USA
○ Mentors: Professors [Kevin R. Coombes](#), [Amanda E. Toland](#), [Sebastian Kurtsek](#), [Tony Nance](#).

Academic Appointments

- 2021–present **Assistant Professor**, [Department of Biostatistics](#) and [Department of Biomedical Informatics](#), Vanderbilt University Medical Center, Nashville, TN, USA

Professional Activities

Intramural

Committees

- 2021–2024 Member of Organizing Committee, *Summer Internship Program for Underrepresented Undergraduates*, Vanderbilt University Medical Center
- 2023–present Member of Admissions Committee, *Biostatistics Graduate Program*, Vanderbilt University

Student Advising

- 2021–2023 Project Advisor, *Yuting Mei*, (Master's student), Data Science, Vanderbilt University
- 2021–2023 Project Advisor, *Chiara Di Gravio*, (Ph.D. student), Biostatistics, Vanderbilt University
- 2022–2023 Thesis Advisor, *Jared Strauch*, (Master's student), Biostatistics, Vanderbilt University
- 2022–2023 Thesis Advisor, *Elisa Yazdani*, (Master's student), Biostatistics, Vanderbilt University
- 2021–2023 Thesis Co-Advisor, *Dora Obodo*, (Ph.D. student), Quantitative & Chemical Biology, Vanderbilt University
- 2023–2024 Project Advisor, *Hao Wang*, (Ph.D. student), Biostatistics, Vanderbilt University

Mentoring

- 2019 Undergraduate Research Mentor (**2** students), REU Program, The Ohio State University
- 2020 Undergraduate Research Mentor (**1** student), Herchel Smith-Harvard Undergraduate Science Research Program, The Ohio State University
- 2022 Undergraduate Research Mentor (**2** students), Summer Internship Program for Underrepresented Undergraduates, Vanderbilt University Medical Center
- 2023 Undergraduate Research Mentor (**2** students), Summer Internship Program for Underrepresented Undergraduates, Vanderbilt University Medical Center

- 2023-2025 Postdoctoral Researcher Mentor, *Leila Azinfar*, R00 HG011367-02 Project, Vanderbilt University Medical Center
- 2024 Undergraduate Research Mentor (1 student), Summer Internship Program for Underrepresented Undergraduates, Vanderbilt University Medical Center
- 2025-present Postdoctoral Researcher Mentor, *Samhita Pal*, ME-2023C1-32148 Project, Vanderbilt University Medical Center

Extramural

Editorial Responsibilities

- 2025-present TOPML: ACM Transactions on Probabilistic Machine Learning *Associate Editor*

Session Organizer

- 2022 Vanderbilt Biostatistics Seminar Series
- 2022 CMStatistics2022 session on “Causality and Machine Learning”

Reviewer

- BMC Bioinformatics (8)
- BMC Genomics (2)
- PLOS Computational Biology (1)
- PLOS Genetics (1)
- Biometrics (1)
- Nature Communications (1)
- Nucleic Acids Research (1)
- Biostatistics (1)
- International Conference on Machine Learning, ICML (4)
- Conference on Uncertainty in Artificial Intelligence, UAI (6)
- Conference on Neural Information Processing Systems, NeurIPS (4)
- Conference on Causal Learning and Reasoning, CLearR (3)

Others

Professional Development

- 2010 Preparing Future Faculty Practicum Course, University of Minnesota
- 2017-2021 Professional Development Seminars, Mathematical Biosciences Institute, The Ohio State University
- 2023-2025 The Geoffrey Fleming Academy for Excellence in Education Mentorship Program (The Fleming Society), Vanderbilt University Medical Center

Panels

- 2018 Panelist at *Career Path Panel* of Summer REU Program: Mathematical Biology Bootcamp, Mathematical Bioscience Institute, The Ohio State University
- 2018 Curriculum development for *Machine Learning for Bioinformatics* course, Department of Biomedical Informatics, The Ohio State University
- 2019 Judge at the *Edward F. Hayes Forum*, The Ohio State University
- 2019 Panelist at *Career Path Panel* of Summer REU Program: Mathematical Biology Bootcamp, Mathematical Bioscience Institute, The Ohio State University
- 2023 Panelist at *Winner Panel* of Bias Detection Tools for Clinical Decision-Making Challenge, NIH's NCAT

- 2024 Panelist at *AI Discovery and Research Panel* of the launch event of AI Discovery & Vigilance to Accelerate Innovation & Clinical Excellence (ADVANCE) Center, Department of Biomedical Informatics, Vanderbilt University Medical Center

Workshop & Conference Participation

- 2011 Oct. IMA Workshop on Large Graphs: Modeling, Algorithms, and Applications, Minneapolis, MN
- 2012 Mar. IMA Workshop on Machine Learning: Theory and Computation, Minneapolis, MN
- 2012 Oct. 21st Conference on Information and Knowledge Management, Maui, HI
- 2015 Feb. IMA Workshop on Convexity and Optimization: Theory and Applications, Minneapolis, MN
- 2015 Apr. Analytic Tools in Probability and Applications, Minneapolis, MN
- 2015 Apr. IMA Workshop on Information Theory and Concentration Phenomena, Minneapolis, MN
- 2015 May. IMA Workshop on Graphical Models, Statistical Inference, and Alg., Minneapolis, MN
- 2015 Jul. 31st Conference on Uncertainty in Artificial Intelligence, Amsterdam, Netherlands
- 2018 May. Stanford Conference on Big Data in Precision Health, Stanford, CA
- 2018 Aug. 34th Conference on Uncertainty in Artificial Intelligence, Monterey, CA
- 2018 Jul. CBMS Conference on Elastic Functional and Shape Data Analysis (EFSDA), Columbus, OH
- 2018 Oct. Nature Conference on Big Data and Cancer Precision Medicine, Boston, MA
- 2018 Oct. MBI Workshop on Math and the Microbiome, Columbus, OH
- 2019 Jun. MBI Workshop on Bayesian Causal Inference, Columbus, OH
- 2019 Jun. 36th International Conference on Machine Learning, Long Beach, CA
- 2019 Jun. Computational Biology Workshop at ICML, Long Beach, CA
- 2019 Jun. Adaptive & Multitask Learning: Algorithms & Sys. Workshop at ICML, Long Beach, CA
- 2019 Nov. MBI Workshop on Evolutionary Dynamics in Cancer, Columbus, OH
- 2020 Jun. 37th International Conference on Machine Learning, Virtual
- 2020 Jun. Computational Biology Workshop at ICML, Virtual
- 2020 Oct. American Society of Human Genetics, Virtual
- 2020 Dec. 34th Conference on Neural Information Processing Systems, Virtual
- 2021 Mar. Network Biology, Cold Spring Harbor Laboratory, Virtual
- 2021 Apr. 24th International Conference on Artificial Intelligence and Statistics, Virtual
- 2021 Apr. Probabilistic Modeling In Genomics, Cold Spring Harbor Laboratory, Virtual
- 2021 Apr. American Association for Cancer Research Annual Meeting, Virtual
- 2021 Jul. 26th Summer Institute in Statistical Genetics, Department of Biostatistics, University of Washington, Virtual
- 2021 Jul. 38th International Conference on Machine Learning, Virtual
- 2021 Jul. 37th Conference on Uncertainty in Artificial Intelligence, Virtual
- 2021 Dec. 35th Conference on Neural Information Processing Systems, Virtual
- 2022 Jul. 27th Summer Institute in Statistical Genetics, Department of Biostatistics, University of Washington, Virtual
- 2022 Dec. 36th Conference on Neural Information Processing Systems, Virtual
- 2022 Dec. 15th International Conference on Computational and Methodological Statistics, Virtual
- 2023 Mar. Network Biology, Cold Spring Harbor Laboratory, Virtual
- 2023 Mar. Probabilistic Modeling in Genomics, Cold Spring Harbor Laboratory, Virtual
- 2023 Mar. Eastern North American Region (ENAR) Meeting, Nashville, TN
- 2023 Apr. Systems Immunology, Cold Spring Harbor Laboratory, Virtual
- 2023 Dec. 37th Conference on Neural Information Processing Systems, Virtual

- 2025 Aug. Joint Statistical Meetings (JSM), Nashville, TN
- 2026 Apr. 5th Conference on Causal Learning and Reasoning (CLearR), Broad Institute, Boston, MA
- 2026 Aug. 42nd Conference on Uncertainty in Artificial Intelligence (UAI), Amsterdam, Netherlands, Virtual

Teaching Experience

The Ohio State University

- 2018 Machine Learning for Bioinformatics *Co-teaching at the Biomedical Informatics Department*
- 2019 Artificial Intelligence II *Teaching at the Computer Science & Engineering Department*
- 2020 Artificial Intelligence II *Teaching at the Computer Science & Engineering Department*

Vanderbilt University

- 2024–2026 BIOS 7362/8365: Advanced Statistical Inference *Spring 2024, 2025, and 2026, Department of Biostatistics*
- 2026 BIOS 7365: Probabilistic Machine Learning and Statistical Modeling *Fall 2026, Department of Biostatistics*

Research Program

- 2020 **K99 HG011367-01 (PI: Asiaee)**, *Causal Effect Estimation of Regulatory Molecules*, NHGRI, \$100k
08/20/2020 - 07/31/2021
- 2021 **R00 HG011367-02 (PI: Asiaee)**, *Causal Effect Estimation of Regulatory Molecules*, NHGRI, \$750k
08/01/2021 - 07/31/2025
- 2024 **ME-2023C1-32148 (PI: Asiaee, Double PI: Huling)**, *Improving Heterogeneous Effect Estimation by Integration of Experimental and Observational Studies*, PCORI, \$750k
02/01/2024 - 01/31/2027
- 2026 **MISM Pilot Award (PI: Asiaee)**, *Multiscale Causal Bridge from Antigen-Perturbed Single Cells to Cytometry Vaccine Biomarkers*, Center of Excellence for Multiscale Immune System Modeling (MISM), NIAID U54-AI191253, \$60k
06/01/2026 - 05/31/2027

Publications and Presentations

Peer-reviewed Articles

Amir Asiaee T., Mariano Tepper, Arindam Banerjee, and Guillermo Sapiro. If you are happy and you know it... tweet. In *21st ACM international conference on Information and knowledge management (CIKM)*, pages 1602–1606. ACM, 2012.

Amir Asiaee T., Mohammad Afshar, and Masoud Asadpour. Influence maximization for informed agents in collective behavior. In *Distributed Autonomous Robotic Systems*, pages 389–402. Springer, 2013.

Golshan Golnari*, **Amir Asiaee T.***, Arindam Banerjee, and Zhi-Li Zhang. Revisiting non-progressive influence models: Scalable influence maximization in social networks. In *31st Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 316–325, 2015 (*Equal contribution).

Amir Asiaee T., Soumyadeep Chatterjee, and Arindam Banerjee. High dimensional structured estimation with noisy designs. In *16th SIAM International Conference on Data Mining (SDM)*, pages 801–809. SIAM, 2016.

Amir Asiaee T., Hardik Goel, Shalini Ghosh, Vinod Yegneswaran, and Arindam Banerjee. Time series deinterleaving of dns traffic. In *1st Deep Learning and Security Workshop*, 2018.

Zachary B. Abrams, Mark Zucker, Min Wang, **Amir Asiaee Taheri**, Lynne V. Abruzzo, and Kevin R. Coombes. Thirty biologically interpretable clusters of transcription factors distinguish cancer type. *BMC Genomics*, 19(1):738, Oct 2018.

Min Ho* Cho, **Amir Asiaee***, and Sebastian Kurtek. Elastic statistical shape analysis of biological structures with case studies: A tutorial. *Bulletin of Mathematical Biology*, 81(7):2052–2073, 2019.

Amir Asiaee, Samet Oymak, Kevin R. Coombes, and Arindam Banerjee. Data enrichment: Multi-task learning in high dimension with theoretical guarantees. In *Adaptive and Multi-Task Learning Workshop at ICML*, 2019.

Amir Asiaee*, Zachary B. Abrams*, Samantha Nakayiza, Deepa Sampath, and Kevin R. Coombes. Explaining gene expression using twenty-one micrnas. *Journal of Computational Biology*, Forthcoming, 2020 (*Equal contribution).

Sadegh Akhundzadeh, Alireza Omid, Zainab Maleki, Amanda E Toland, Kevin R Coombes, and **Amir Asiaee**. Learning cancer progression network from mutation allele frequencies. In *Computational Biology Workshop at ICML*, 2020.

Phillip B Nicol, Kevin R Coombes, Courtney Deaver, Oksana Chkrebti, Subhadeep Paul, Amanda E Toland, and **Amir Asiaee**. Oncogenetic network estimation with disjunctive bayesian networks. *Computational and Systems Oncology*, 1(2):e1027, 2021.

Phillip B. Nicol, Dániel L. Barabási, Kevin R. Coombes, and **Amir Asiaee**. SITH: An R package for visualizing and analyzing a spatial model of intratumor heterogeneity. *Computational and Systems Oncology*, 2(2):e1033, 2022.

Amir Asiaee, Zachary B Abrams, Heather H Pua, and Kevin R Coombes. Transcriptome complexity disentangled: A regulatory molecules approach. *International Journal of Molecular Sciences*, 26(6):2510, 2025.

Hongxu Zhu, **Amir Asiaee**, Leila Azinfar, Jun Li, Han Liang, Ehsan Irajizad, Kim-Anh Do, and James P Long. AUPRC: a metric for evaluating the performance of in-silico perturbation methods in identifying differentially expressed genes. *Briefings in Bioinformatics*, 26(5):bbaf426, September 2025.

Eric Milan Mukherjee, Dodie Park, **Amir Asiaee**, Matthew S. Krantz, Cosby A. Stone Jr, Michelle Martin-Pozo, and Elizabeth Phillips. Immune checkpoint inhibitors as independent and synergistic drivers of SJS/TEN: An analysis of FAERS. *JAMA Oncology*, 11(12):1542–1545, 2025.

Amir Asiaee, Samhita Pal, Cole Beck, and Jared D. Huling. Sharp bounds for treatment effect generalization under outcome distribution shift. In *Proceedings of the 5th Conference on Causal Learning and Reasoning (CLear)*, volume 323 of *Proceedings of Machine Learning Research*, pages 1572–1603, 2026.

Zhuohui J. Liang, Zhuohang Li, Nicholas J. Jackson, Kevin H. Guo, Yanink Caro-Vega, Ronaldo I. Moreira, Fabio Paredes, Jordany Bernadin, Diana Varela, Carina Cesar, Alessandro Blasimme, Jessica M. Perkins, **Amir Asiaee**, Stephany N. Duda, Bradley A. Malin, Bryan E. Shepherd, and Chao Yan. Generating synthetic multi-national longitudinal cohorts for clinically grounded HIV research. *Nature Communications*, 17(1):7857, 2026.

Thanaporn Ratchataswan, Rebecca Lee, **Amir Asiaee**, Matthew S. Krantz, Grace Koo, Cosby A. Stone, and Elizabeth J. Phillips. Multiple antibiotic allergy evaluation strategy (MAAES): A comparative effectiveness study. *The Journal of Allergy and Clinical Immunology: In Practice*, 2026. In press.

Simon Mallal, **Amir Asiaee**, and Elizabeth Phillips. Keystone epitope theory: Implications for hypersensitivity, autoimmunity, and transplantation. *Pathogens and Immunity*, 11(2):16–38, 2026.

Simon Mallal and **Amir Asiaee**. Keystone epitope theory: An ecological perspective on RNA viruses, tumor immunoediting, and vaccine design. *Pathogens and Immunity*, 11(2):39–65, 2026.

Amir Asiaee and Kaveh Aryan. Workload-preserving differentially private synthetic data for causal inference via maximum-entropy calibration. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI)*, volume 337 of *Proceedings of Machine Learning Research*, pages 133–160, 2026.

Amir Asiaee, Kaveh Aryan, and James P. Long. Partial causal structure learning for valid selective conformal inference under interventions. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI)*, volume 337 of *Proceedings of Machine Learning Research*, pages 161–184, 2026. Oral presentation.

Amir Asiaee and Samhita Pal. Improving RCT-based treatment effect estimation under covariate mismatch via calibrated alignment. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI)*, volume 337 of *Proceedings of Machine Learning Research*, pages 185–209, 2026.

Amir Asiaee. Certified interventional fidelity: Anytime-valid, adaptive evaluation of causal claims in mechanistic interpretability. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI)*, volume 337 of *Proceedings of Machine Learning Research*, pages 210–227, 2026.

Presentations

- 2018 Mar. *Generalized High Dimensional Data Sharing with Application*, MBI Seminar, Columbus, OH, USA
- 2018 Oct. *Inferring Mutations Order from Cross-sectional Cancer Data*, MBI Seminar, Columbus, OH, USA
- 2019 Nov. *Disjunctive Bayesian Network Infers Cancer Progression Network*, Evolutionary Dynamics in Cancer Workshop, MBI, Columbus, OH, USA
- 2019 Nov. *Searching for Effective Combination Cancer Therapy Using Single Doses*, MBI Seminar, Columbus, OH, USA
- 2020 Feb. *Disjunctive Bayesian Network Infers Cancer Progression Network*, Department of Biostatistics, Vanderbilt University, Nashville, TN, USA
- 2021 Oct. *Whole Transcriptome Prediction from Regulatory Molecules*, Bioinformatics Working Group, Department of Biostatistics, Vanderbilt University, Virtual
- 2022 Feb. *Identification and Comparison of Genes Differentially Regulated by Transcription Factors and MicroRNAs*, Department of Biostatistics, University of Louisville, Virtual
- 2022 Dec. *Methods to improve the efficiency of RCTs using observational studies*, Invited Session (Combining Clinical Trials and Observational Study Data) at International Conference on Computational and Methodological Statistics, Virtual
- 2024 Jun. *Leveraging Observational Data for Efficient CATE Estimation in Randomized Controlled Trials*, Invited Session (Advancing Clinical Trial Analysis) at International Chinese Statistical Association, Nashville, TN, USA

- 2025 Jul. *Combining Observational and RCT for Heterogeneous Treatment Effect Estimation*, Statistical Innovation in Clinical Trials Meeting, Department of Biostatistics, Vanderbilt University Medical Center, Nashville, TN, USA
- 2026 Apr. *Data Borrowing from Observational Studies and Generalizing RCTs*, Causal Inference Working Group, Department of Biostatistics, Vanderbilt University Medical Center, Nashville, TN, USA
- 2026 Aug. *Partial Causal Structure Learning for Valid Selective Conformal Inference under Interventions*, Oral Presentation, 42nd Conference on Uncertainty in Artificial Intelligence (UAI), Amsterdam, Netherlands, Virtual
- 2026 Aug. *Predicting Cellular Responses to Unseen Perturbations: Misleading Metrics and Missing Uncertainty*, Invited Seminar, Department of Biostatistics, Data Science, and Epidemiology, Georgia Cancer Center, Augusta University, Augusta, GA, USA

Abstracts and Posters

- 2012 Oct. *If You are Happy and You Know It... Tweet*, 21st Conference on Information and Knowledge Management, Maui, HI
- 2015 Jul. *Revisiting Non-progressive Influence Models: Scalable Influence Maximization in Social Networks*, 31st Conference on Uncertainty in Artificial Intelligence, Amsterdam, Netherlands
- 2018 May. *Precision Cancer Treatment with High Dimensional Data Sharing Model*, Stanford Conference on Big Data in Precision Health, Stanford, CA
- 2018 Oct. *Precision Cancer Treatment with High Dimensional Data Enrichment*, Nature Conference on Big Data and Cancer Precision Medicine, Boston, MA
- 2019 Jun. *Explaining Gene Expression Using Twenty-One MicroRNAs*, Computational Biology Workshop at ICML, Long Beach, CA
- 2019 Jun. *Data Enrichment: Multi-task Learning in High Dimension with Theoretical Guarantees*, Adaptive and Multi-Task Learning Workshop at ICML, Long Beach, CA
- 2020 Jun. *Learning Cancer Progression Network from Mutation Allele Frequencies*, Computational Biology Workshop at ICML, Virtual
- 2021 Apr. *Identification and Comparison of Genes Differentially Regulated by Transcription Factors and MicroRNAs*, NHGRI Research Training and Career Development Annual Meeting, Virtual
- 2026 Apr. *Sharp Bounds for Treatment Effect Generalization under Outcome Distribution Shift*, 5th Conference on Causal Learning and Reasoning (CLear), Broad Institute, Boston, MA
- 2026 Aug. *Workload-Preserving Differentially Private Synthetic Data for Causal Inference via Maximum-Entropy Calibration*, 42nd Conference on Uncertainty in Artificial Intelligence (UAI), Virtual
- 2026 Aug. *Improving RCT-Based Treatment Effect Estimation Under Covariate Mismatch via Calibrated Alignment*, 42nd Conference on Uncertainty in Artificial Intelligence (UAI), Virtual
- 2026 Aug. *Certified Interventional Fidelity: Anytime-Valid, Adaptive Evaluation of Causal Claims in Mechanistic Interpretability*, 42nd Conference on Uncertainty in Artificial Intelligence (UAI), Virtual